

NOA PhD Dissertation — Integration Report

Output file: [path]
Final size: 4,213,371 bytes (4.02 MB)

Sections Added

#	Section ID	Title	Inserted After
1	literature-expanded	Expanded Literature Review	#literature
2	preprocessing-fe	Data Preprocessing & Feature Engineering	#data-prep
3	training-experiments	Model Training & Experiments	#performance
4	validation-methods	Validation Methods	#training-experiments
5	xai-pipeline	XAI Pipeline (Phase 5)	#xai

Content Integrated

- **Preprocessing** — full dataset info (2450 patients, 55→73 features), 10 cleaning steps, full feature-engineering breakdown, 7 transformations, outcome analysis.
- **Training** — 9-classifier benchmark, hyperparameter tuning (Optuna), 14 experiment logs, 5 ablation studies, 5-version model evolution (Paper 1 → Paper B), 20 decisions, 15 problems/solutions, 8 code-evolution phases.
- **Validation** — 10 validation methods, 15 base models (27 including adjusted variants), full performance/calibration/threshold/risk-stratification/subgroup analyses, TRIPOD/PROBAST/STROBE compliance.
- **XAI Pipeline** — 10-task pipeline, 15 interpretability methods, complete SHAP rankings, 7 LIME archetypes, counterfactuals, 5 ablation studies, 430+ SHAP/LIME visualizations across all models and archetypes visualizations.
- **Expanded Literature** — 41 references, 7 research gaps, 9 theoretical frameworks, 21 methodologies, 8 narrative sections, full citation map.

Logos Updated

- **MUMS Logo** — embedded as base64 (JPEG) — added to cover header & footer
- **Medical Informatics Logo** — embedded as base64 (JPEG) — added to cover header & footer
- **Royan Institute Logo** — embedded as base64 (PNG) — kept in cover header & footer

All three logos appear in the cover page header (`.logos` block) and in a new persistent footer that renders below every page section.

Issues Fixed

- ☒ “Jamalirad” used consistently (single word, normalized via regex).
- ☒ No occurrence of “Farnaz Khoshrounejad” (verified absent in source template — none added).
- ☒ Dr. Vakili-Arki preserved in cover meta cards, supervisor JS object, and floating profile cards.
- ☒ All forbidden platform-name references stripped (regex pass).
- ☒ Sticky ToC updated with the 5 new section anchors and new groupings.
- ☒ Color scheme (`#0f6d8b` , `#2856a8`), Inter typography, sticky TOC, search, lightbox, supervisor floating cards — all preserved.
- ☒ Existing CSS classes reused for new sections (`.section-title` , `.flow-step` , `.flow-steps` , `.kpi` , `details` , `.pill` , `.topic-accordion` , `.two-col` , `.footer-note` , etc.).
- ☒ All existing JavaScript (Chart.js initializers, search, sorting, mind-map, lightbox, supervisor hover) untouched.

Quality Checks

- HTML validates structurally (single `<main>` , balanced `<section>` blocks).
- Internal anchor links from the new ToC entries resolve to the new section IDs.
- Logos embedded as base64 — file is fully self-contained, no external image dependencies for branding.
- Final size: **4,213,371 bytes**.